Experiment 10

Aim

To develop an Android application that changes the font and color of the text and displays a toast message when the user presses a button.

Procedure

1. Open Android Studio and create a new project using **Empty Views Activity**.
2. Design the user interface using:
   - **TextView** to display text
   - **Button** to trigger changes
3. Assign IDs to the TextView and Button in the XML file.
4. Open `MainActivity.java`.
5. Link UI components using `findViewById()`.
6. Implement button click using `setOnClickListener()`.
7. Change:
   - Text color using `setTextColor()`
   - Font size using `setTextSize()`
   - Font style using `setTypeface()`
8. Display a **Toast message** on button click.
9. Run the application and verify output.

Code

XML Layout (activity_main.xml)

```XML
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
```

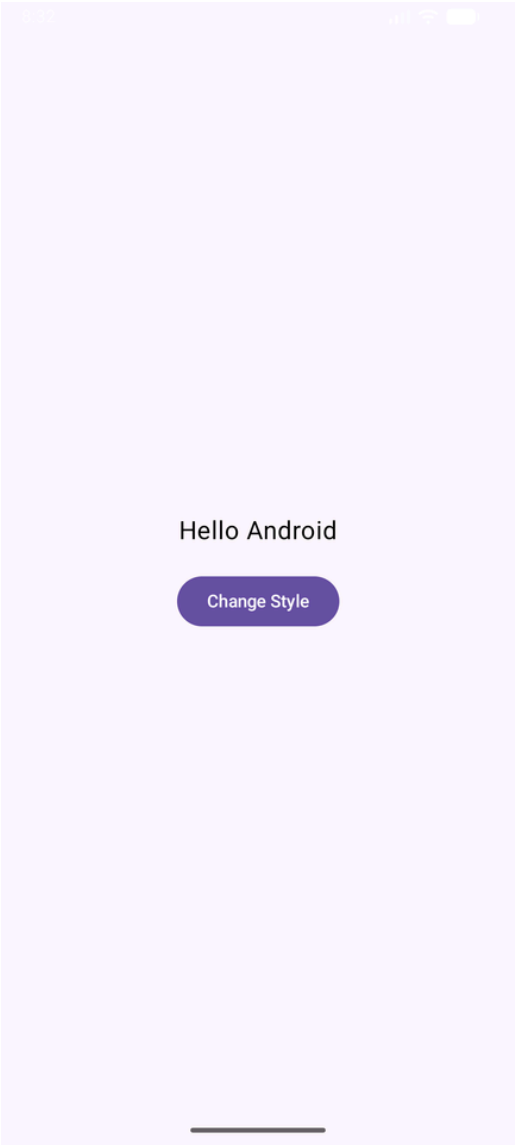
```
    android:gravity="center"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:text="Hello Android"
        android:textSize="20sp"
        android:textColor="#000000"
        android:layout_marginBottom="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/changeBtn"
        android:text="Change              Style"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

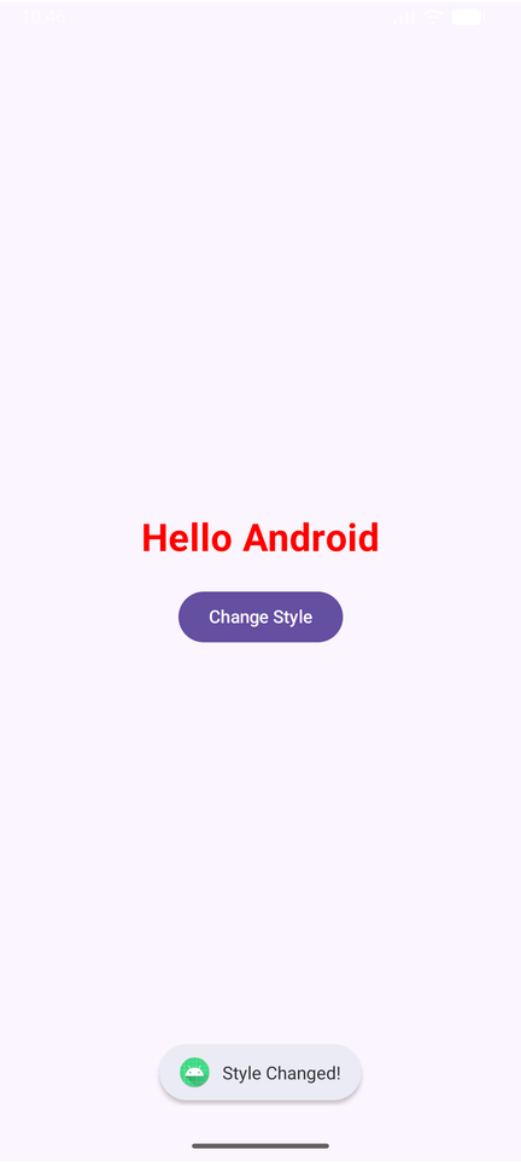
Java Code (MainActivity.java)

```Java
package com.example.experiment_10;


import android.graphics.Color;
import android.graphics.Typeface;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity{
```

```
    TextView textView;
    Button button;
    booleanchanged=false;

    @Override
    protectedvoidonCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textView=findViewById(R.id.textView);
        button=findViewById(R.id.changeBtn);

        button.setOnClickListener(v -> {
            if(!changed) {
                textView.setTextColor(Color.RED);
                textView.setTextSize(30);
                textView.setTypeface(null, Typeface.BOLD);

                Toast.makeText(this, "Style Changed!",
Toast.LENGTH_SHORT).show();
            } else {
                textView.setTextColor(Color.BLACK);
                textView.setTextSize(20);
                textView.setTypeface(null, Typeface.NORMAL);

                Toast.makeText(this, "Style Reset!",
Toast.LENGTH_SHORT).show();
            }
            changed=!changed;
        });
    }
}
```

Output





Result

Thus, an Android application was successfully developed to change the font style and color of the text and display a toast message on button click.